

GPU Performance — Compute-It-Yourself

Use the Formula Sheet equations to compute real numbers about your GPU — each answer hands you an insight a spec sheet won't.

SYMBOL	MEANING	EXAMPLE — RTX 5090 (USE YOURS)
C	VRAM capacity	32 GB
BW	memory bandwidth	1790 GB/s
P_{peak}	measured peak FP16 compute (Lab C)	~210 TFLOP/s
W_{pow}	board power	~575 W

A · Warm-ups (Q1–3) · **B · Throughput, batching & fit** (Q4–8) · **C · Multi-GPU & efficiency** (Q9–11). Write an insight for each.

Q1. Decode is bandwidth, not FLOPs.

13B model, FP16. At batch 1, decode re-reads *all* weights every token.

Find: (a) weight bytes/token; (b) decode tok/s $\approx$ BW $\div$ weight GB.

INSIGHT

Q6. KV cache vs weights.

13B, 40 layers, hidden 5120, MHA: KV/token = $2 \times \text{layers} \times \text{hidden} \times 2 \text{ B}$.

Find: KV/token, then total at **S = 32k** (batch 1). Compare to 26 GB of weights.

INSIGHT

Q2. Arithmetic intensity of decode.

Matrix $\times$ vector: FLOPs = $2n^2$, bytes $\approx 2n^2$.

Find: AI = FLOPs $\div$ bytes. Does it depend on n ?

INSIGHT

Q7. Prefill vs decode — which wait dominates?

Prompt 2000, generate 500. Prefill $\approx$ 8000 tok/s; decode = your Q1.

Find: prefill time & decode time. Which is bigger?

INSIGHT

Q3. Your GPU's ridge point.

Find: Ridge = P_{peak} $\div$ BW. Classify decode (AI $\approx$ 1) & prefill at $n=4096$ (AI $\approx$ $n/2$).

INSIGHT

Q8. Will a 34B fit? (with KV)

32 GB card, 34B, context 8k, batch 4, KV $\approx$ 0.2 MB/token (GQA).

Find: weights FP16 & INT4; KV; total in INT4 (+2 GB). Fit?

INSIGHT

Q4. The batching sweet spot.

A batch of **B** decode requests has intensity AI $\approx$ B.

Find: how many concurrent requests **B** to reach your ridge (compute-bound)?

INSIGHT

Q9. The no-NVLink tax.

TP across 2 GPUs over PCIe (50 GB/s). Per layer bytes = $1 \times 2048 \times 4096 \times 2$, 32 layers.

Find: comm time per layer, then per token ($\times 32$).

INSIGHT

Q5. The quantization double-win.

Same 13B, FP16 vs INT4.

Find: (a) weight GB each; (b) decode tok/s each; (c) speedup.

INSIGHT

Q10. Two 5090s: split it or double it?

13B fits on **one** 32 GB card. $2 \times$ 5090, **no NVLink**.

Find: (A) tensor-split (Q9 tax/token) vs (B) data-parallel aggregate tok/s?

INSIGHT

Q11. Tokens per watt (stretch).

69 tok/s at 575 W.

Find: tokens/joule; energy per 1M tokens (kWh); cost @ \$0.05/kWh. Effect of batching?

INSIGHT

Companion to the Tasar'u GPU Infrastructure Labs · swap in your own measured numbers — the insights hold, only the magnitudes change.